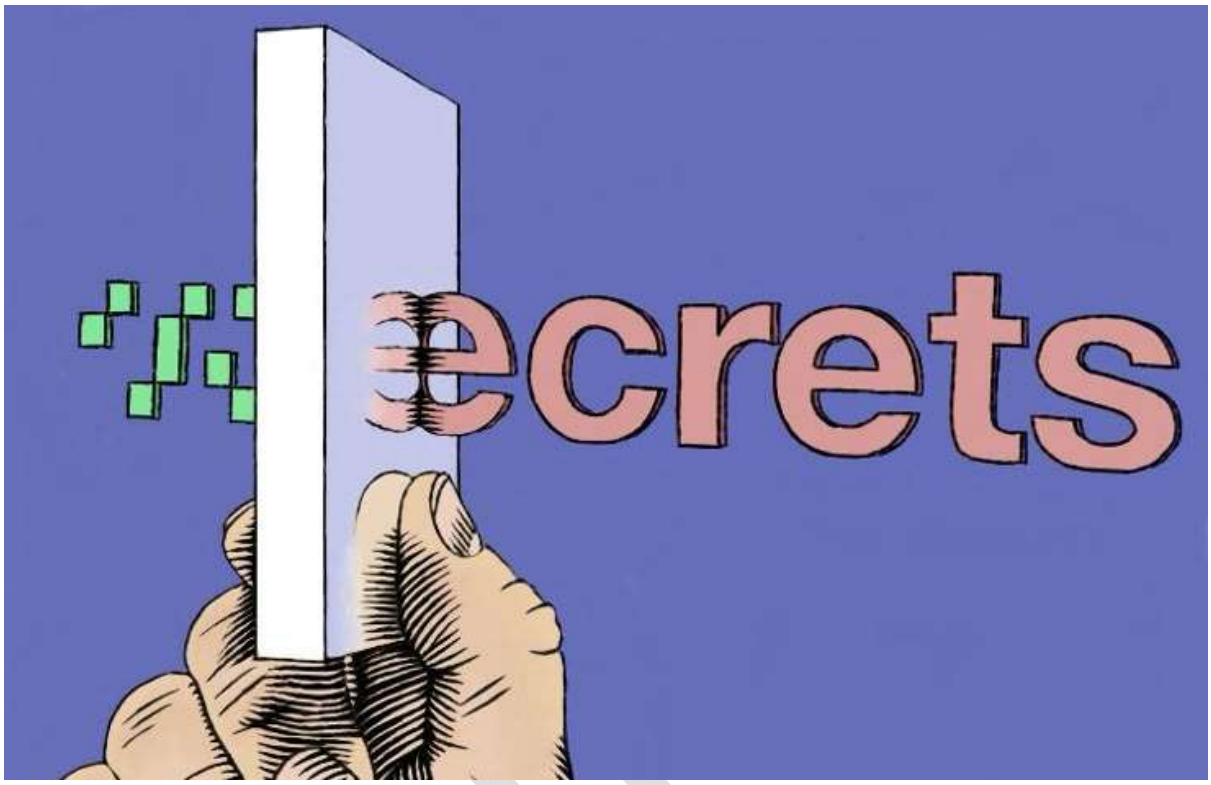
Cryptography (Part-II)

Basics



Public Key Cryptography builds upon foundational cryptographic principles to enable secure communication over insecure networks by introducing asymmetric key mechanisms, where a public key is used for encryption and a private key for decryption. It ensures critical security properties such as authentication, integrity, authenticity, and confidentiality, making it essential for modern digital systems. Core technologies like RSA and Diffie-Hellman enable secure key exchange and encryption, while real-world implementations such as SSH, SSL/TLS certificates, and GPG demonstrate how these concepts are applied in secure remote access, web communication, and encrypted messaging.

This documentation and learning approach is strongly inspired by the structured, hands-on content provided by TryHackMe, which significantly contributed to practical understanding of Public Key Cryptography Basics concepts.

DOCUMENTED BY

Kaustubh Ghosalkar aka "WEIRDO"

LINKEDIN GITHUB

Start!!!

1. Core Security Principles (CIA + AAA Model)

Principle	Meaning	Example
Authentication	Verifying identity	SSH login
Authenticity	Message is from real sender	Digital signature
Integrity	Data not altered	Hash verification
Confidentiality	Only authorized access	Encryption (TLS/SSH)

2. Asymmetric Encryption (Public Key Cryptography)

📌 **Concept**

- Uses **2 keys**:
  - **Public Key → Encrypt**
  - **Private Key → Decrypt**

📌 **Real-World Workflow**

1. Server shares **public key**
2. Client encrypts data using it
3. Server decrypts using **private key**

🔐 **Key Insight**

- Used for:
  - Secure key exchange
  - Authentication
  - Digital signatures

WEIRDO

3. RSA Algorithm (Core Concept + Variables)

📌 **Important Variables**

Variable	Meaning
p, q	Prime numbers
n = p × q	Modulus
φ(n)	Euler's Totient
e	Public exponent
d	Private exponent
m	Plaintext
c	Ciphertext

📌 **RSA Workflow (Simplified)**

```
# Encryption
c = m^e mod n

# Decryption
m = c^d mod n
```

📌 **Practical Example**

p = 4391
q = 6659

n = p * q = 29239669
φ(n) = (p-1)(q-1) = 29228620

⚠️ **Key Insight**

- Security based on:
  Difficulty of factorization of large numbers

WEIRDO

4. Diffie-Hellman Key Exchange

📌 **Purpose**

Securely share a **secret key over insecure channel**

📌 **Formula Flow**

A = g^a mod p
B = g^b mod p

Shared Key:
K = B^a mod p = A^b mod p

📌 **Example**

p = 29
g = 5

a = 12 → A = 7
b = 17 → B = 9

Shared Key = 24

⚠️ **Key Insight**

- Used in:
  - TLS Handshake
  - Secure sessions

WEIRDO

5. SSH (Secure Shell)

📌 Purpose

Secure remote access using encryption + authentication

🔧 Key Commands

```
# Connect to server
ssh user@ip_address

# Generate key pair
ssh-keygen -t ed25519

# Use private key
ssh -i keyfile user@host
```

📌 Important Concepts

- Public key stored in:

~/.ssh/authorized_keys

- Private key permissions:

chmod 600 private_key

⚠️ Security Notes

- Never share private keys
- Use passphrase for protection

WEIRDO

6. Digital Signatures & Certificates

📌 Digital Signature Workflow

1. Hash message
2. Encrypt hash using private key
3. Receiver decrypts using public key
4. Compare hashes → Verify integrity

📌 Certificates (HTTPS)

- Used in **SSL/TLS**
- Issued by **Certificate Authorities (CA)**

🔧 Example Tools / Services

- Free TLS Certificate:

Let's Encrypt

⚠️ Key Insight

- Provides:
    - Authentication
    - Integrity
    - Trust chain

WEIRDO

7. PGP & GPG (Encryption + Signing)

📌 **Purpose**

- Secure emails
- File encryption
- Digital signing

🔧 **Commands**

```
# Generate key
gpg --full-gen-key

# Import key
gpg --import backup.key

# Decrypt file
gpg --decrypt file.gpg
```

📌 **Use Case**

- Encrypt using recipient's **public key**
- Decrypt using own **private key**
-

⚠️ **Security Note**

- Backup keys securely
- Protect with passphrase

WEIRDO

8. Attack Concepts (Very Important)

Attack	Description
Brute Force	Try all combinations
Dictionary Attack	Try common words
Cryptanalysis	Break crypto logic

9. Key Takeaways (Revision Section)

- RSA → Secure encryption via factorization problem
- Diffie-Hellman → Secure key exchange
- SSH → Secure remote login using keys
- TLS Certificates → Website identity verification
- GPG → Email/file encryption + signing
- Digital Signatures → Integrity + authenticity

WEIRDO

10. Practical Learning Stack (Your Usage)

🧠 Must-Know Commands

```
ssh-keygen
ssh user@host
gpg --decrypt file.gpg
gpg --import key
```

🧠 Must-Know Terms

- Public Key / Private Key
- Digital Signature
- Certificate Authority
- Key Exchange
- Cipher / Algorithm

✅ Final Insight

This module transitions you from **basic cryptography → real-world implementations**, focusing on how asymmetric encryption powers secure communication protocols used in **SOC operations, penetration testing, and secure infrastructure design**.

WEIRDO